1. Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation (Point 4 in the list)

This is the starting point for solving this problem. Students must carefully read and understand the introduction, tabular data, and the meaning of each statement.

- **Specific Analysis:** Students need to understand key concepts such as "average fault rate" and "randomly selected," and extract specific values from the table (e.g., 2000 video players, 6000 audio players, fault rates of 5% and 3% respectively). Integrating this information allows them to build a mental model of Electrix Company's daily production operations, including total output, category-wise output, and fault probabilities, which is the basis for all subsequent calculations.

2. Identify the mathematical aspects of a real-world problem situation including identifying the significant variables (Point 6 in the list)

After establishing the situational model, students need to translate the specific production scenario into abstract mathematical concepts.

- **Specific Analysis:**
  - **Identify key variables:** Daily video player output $V = 2000$, daily audio player output $A = 6000$, total output $T = V + A$, video fault rate $P_V = 5\%$, and audio fault rate $P_A = 3\%$.
  - **Identify mathematical aspects:** The problem involves proportions (first statement), absolute quantities (second statement), and probability (third statement).

3. Translate a problem into a standard mathematical representation or algorithm (Point 9 in the list)

This ability is reflected in converting each specific statement into a mathematical calculation or logical judgment.

- **Specific Analysis:**
  - For the first statement ("One third of the players produced daily are video players"), students need to translate it into a ratio calculation: compute $\frac{V}{T} = \frac{2000}{8000} = \frac{1}{4}$ and compare the result with $\frac{1}{3}$.
  - For the third statement, it needs to be translated into a probability judgment: identifying and converting the 3% from the table directly into a probability value of 0.03.

4. Construct and communicate explanations and arguments in the context of the problem (Point 7 in the list)

The question asks students to circle "Yes" or "No", which means they need to make a clear judgment on the correctness of each statement, and usually need to show the calculation process to support this judgment (although not explicitly required here, this is part of good mathematical practice).

- **Specific Analysis:** Students must use the above mathematical calculations to draw conclusions, such as that the first statement is "No" (because it is one quarter, not one third), and be prepared to explain why.

1. 阅读、解码和理解陈述、问题、任务、对象或图像以创建情境模型 (列表中的第 4 点)

这是解决此问题的起点。学生必须仔细阅读并理解引言、表格数据和每个陈述的含义。

- **具体分析：** 学生需要理解“平均故障率”、“随机挑选”等关键概念，并从表格中提取具体数值（如视频播放器 2000 台，音频播放器 6000 台，故障率分别为 5% 和 3%）(p. 1)。将这些信息整合起来，在大脑中建立一个 Electrix 公司日常生产运作的心理模型，包括总产量、各类产量和故障概率，这是后续所有计算的基础。

2. 识别现实世界问题情境的数学方面，包括识别重要变量 (列表中的第 6 点)

在建立了情境模型后，学生需要将具体的生产场景转化为抽象的数学概念。

- **具体分析：**
  - **识别重要变量：** 每天的视频播放器产量 $V = 2000$ (p. 1)，每天的音频播放器产量 $A = 6000$ (p. 1)，总产量 $T = V + A$，视频故障率 $P_V = 5\%$ (p. 1)，音频故障率 $P_A = 3\%$ (p. 1)。
  - **识别数学方面：** 问题涉及比例（第一个陈述）、绝对数量（第二个陈述）和概率（第三个陈述）。

3. 将问题转化为标准的数学表示或算法 (列表中的第 9 点)

这一能力体现在将每个具体的陈述转化为数学计算或逻辑判断。

- **具体分析：**
  - 对于第一个陈述“三分之一的播放器是视频播放器”，学生需要将其转化为一个比例计算：计算 $\frac{V}{T} = \frac{2000}{8000} = \frac{1}{4}$，并将结果与 $\frac{1}{3}$ 进行比较。
  - 对于第三个陈述，需要转化为一个概率判断：将表格中的 3% 直接识别并转换为概率值 0.03 (p. 1)。

4. 构建并传达基于问题背景的解释和论证 (列表中的第 7 点)

题目要求学生圈出“Yes”或“No”，这意味着他们需要对每个陈述的正确性做出明确的判断，并且通常需要展示计算过程来支持这个判断（尽管此处未明确要求展示过程，但这是良好数学实践的一部分）。

- **具体分析：** 学生必须通过上述数学计算得出结论，例如第一个陈述是“No”（因为是四分之一而不是三分之一），并准备好解释原因。